Number Systems -- Guided Notes ANSWER KEY

Unit 1.3 | CompTIA Network+ N10-009 Obj. Core 1 (foundational) | N10-008

For instructor use / distribute after students complete the notes.

Question 1

Binary is a base-_______ number system using only the digits _______ and _______. Each position represents a power of two. Fill in the missing position values from left to right:

128 | ___ | ___ | 16 | ___ | 4 | ___ | 1

Answer: Base-2; 0 and 1. Position values: 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1.

Question 2

Convert the decimal number 157 to binary. Use the subtraction method: start at the highest position (128), subtract if it fits, and record a 1; otherwise record a 0. Show your work step by step.

Answer: 128 fits (157 - 128 = 29). 64 does not fit (29 < 64). 32 does not fit (29 < 32). 16 fits (29 - 16 = 13). 8 fits (13 - 8 = 5). 4 fits (5 - 4 = 1). 2 does not fit (1 < 2). 1 fits (1 - 1 = 0). Result: 10011101.

Question 3

Hexadecimal is a base-_______ number system. It uses the digits 0-9 and the letters _______. The letter A = _______, B = _______, C = _______, D = _______, E = _______, F = _______.

Answer: Base-16; A through F. A=10, B=11, C=12, D=13, E=14, F=15.

Question 4

To convert binary to hexadecimal, group the binary digits into sets of _______. For example, convert 11001010 to hex:

1100 = _______ | 1010 = _______

Result in hex: _______

Answer: Groups of 4 (nibbles). 1100 = 12 = C; 1010 = 10 = A. Result: CA (or 0xCA).

Question 5

The hexadecimal value 0xFF equals _______ in decimal. In binary, this is _______ (all eight bits set to 1). This is the maximum value that can be stored in _______ byte of memory.

Answer: 255; 11111111; one (1).

Question 6

Data transfer speeds are measured in _______ (bits per second). Storage sizes are measured in _______ (bytes). Since 1 byte = _______ bits, a 1 Gbps (gigabit per second) connection transfers data at _______ MB/s (megabytes per second).

Answer: Megabits or gigabits (Mbps / Gbps); megabytes or gigabytes (MB / GB); 8 bits; 125 MB/s.

Question 7

The prefix _______ (e.g., 0xFF) indicates a hexadecimal value. The APIPA address range _______._______.x.x is assigned to a device when it cannot reach a DHCP server. A MAC address is a 48-bit address written as _______ hexadecimal pairs (e.g., AA:BB:CC:DD:EE:FF).

Answer: 0x; 169.254; six (6) hexadecimal pairs.

Question 8 -- Real World Application

REAL WORLD: A network technician reads a MAC address filter log and sees the entry: 00:1A:2B:3C:4D:5E. Convert the last octet (5E) from hexadecimal to binary and then to decimal. Explain why hex is used for MAC addresses instead of binary or decimal.

Answer: 5E in hex: 5 = 0101, E = 1110, so 5E = 01011110 in binary. 01011110 in decimal: 64+16+8+4+2 = 94. Hex is used for MAC addresses because it is compact (12 hex characters vs. 48 binary digits) and human-readable. Binary is too long to work with; decimal doesn't divide cleanly across 4-bit groups. Each hex digit maps perfectly to exactly 4 binary bits, making encoding and reading addresses efficient.